

Problem

Show that $\int_{-\infty}^{\infty} \left(\frac{\sin a\omega}{a\omega} \right)^2 d\omega = \frac{\pi}{a}$ Hint: Use the fact that $\mathcal{F}[u(t+a) - u(t-a)] = 2a \left(\frac{\sin a\omega}{a\omega} \right)$

Step-by-step solution

Step 1 of 1

Using Parseval's theorem,

$$\int_{-\infty}^{\infty} f^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega.$$

if $f(t) = u(t+a) - u(t-a)$, then

$$\begin{aligned} \int_{-\infty}^{\infty} f^2(t) dt &= \int_{-a}^a (1)^2 dt = 2a \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 4a^2 \left(\frac{\sin a\omega}{a\omega} \right)^2 d\omega \end{aligned}$$

$$\int_{-\infty}^{\infty} \left(\frac{\sin a\omega}{a\omega} \right)^2 d\omega = \frac{2\pi a}{4a^2} = \frac{\pi}{a} \text{ as required}$$